

A 2D IONOSPHERIC SHELL MODEL AS A BOUNDARY CONDITION FOR A GLOBAL
MHD SPACE WEATHER SIMULATION

by

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Abstract

Acknowledgements

Contents

1	Introduction	1
2	Physical Background	2
3	Conductivity Modelling	3
4	Numerical Implementation	4
4.1	Convergence Study	4
5	Polar Boundary Conditions	7
6	Results	8
7	Conclusions and Future Work	9
	Bibliography	9

List of Figures

4.1	TEST2	4
4.2		5
4.3		5
4.4	Mean radial current processing and overhead for the MueLu	6

List of Tables

Chapter 1

Introduction

Chapter 2

Physical Background

$$j_R(R_E) = [\nabla_\perp \cdot (\Sigma \cdot \nabla \psi)_\perp] \quad (2.1)$$

$$\Sigma_{\text{Amm}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{\Sigma_0 \Sigma_P}{C} & \frac{\Sigma_0 \Sigma_H (-\cos \varepsilon)}{C} \\ 0 & \frac{\Sigma_0 \Sigma_H \cos \varepsilon}{C} & \Sigma_P + \frac{\Sigma_H^2 \sin^2 \varepsilon}{C} \end{pmatrix} \quad (2.2)$$

Chapter 3

Conductivity Modelling

$$\Sigma_P = \frac{40E_0\sqrt{Q}}{16 + E_0^2}, \quad \Sigma_H = 0.45 E_0^{0.85} \Sigma_P \quad (3.1)$$

Chapter 4

Numerical Implementation

4.1 Convergence Study

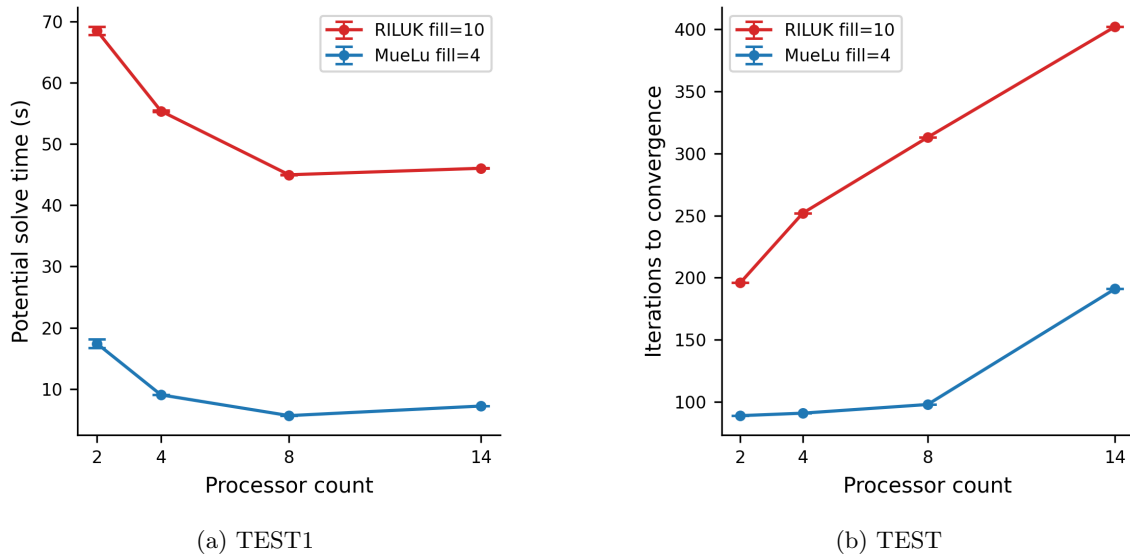


Figure 4.1: TEST2

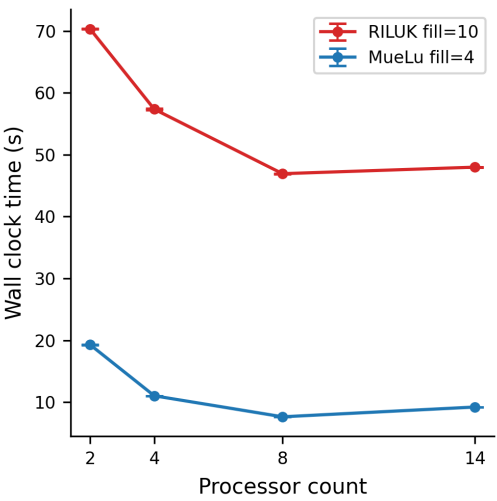


Figure 4.2

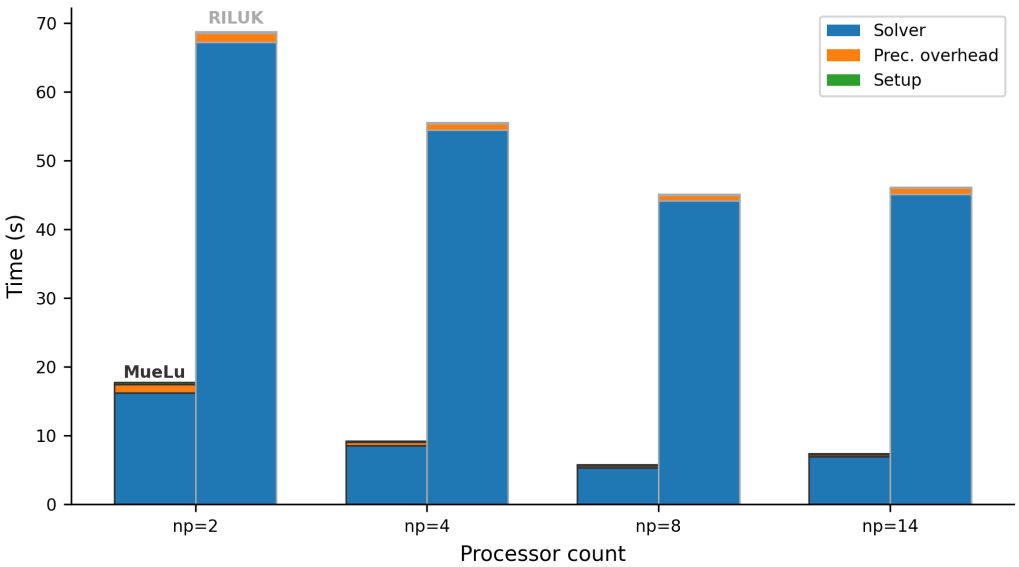


Figure 4.3

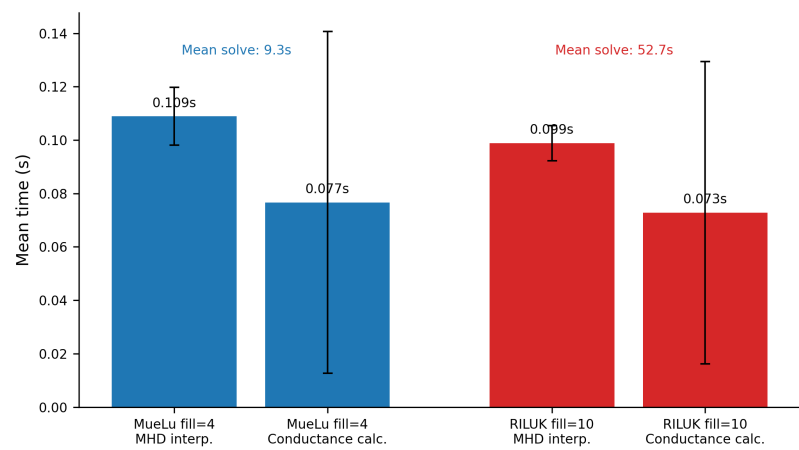


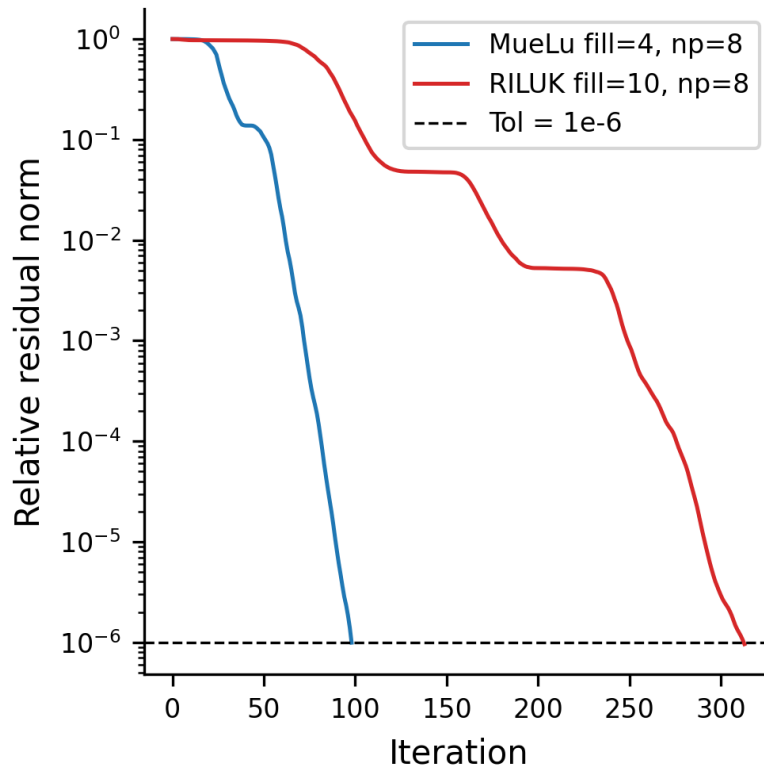
Figure 4.4: Mean radial current processing and overhead for the MueLu

Chapter 5

Polar Boundary Conditions

$$J_{\text{cap}} = \iint_C j_R(R_E) dA = \oint_{\partial C} [(\Sigma \cdot \nabla \psi)_\perp] dl \quad (5.1)$$

$$J_{\text{cap}} = \int_0^{2\pi} \left[\sin \theta_0 \cdot \frac{\partial \psi}{\partial \theta}(\theta_0, \phi) \cdot \Sigma_{\theta\theta}(\theta_0, \phi) + \frac{\partial \psi}{\partial \phi}(\theta_0, \phi) \cdot \Sigma_{\theta\phi}(\theta_0, \phi) \right] d\phi \quad (5.2)$$



Chapter 6

Results

Chapter 7

Conclusions and Future Work

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